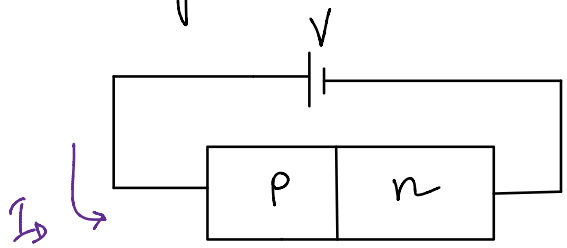
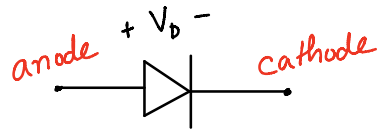
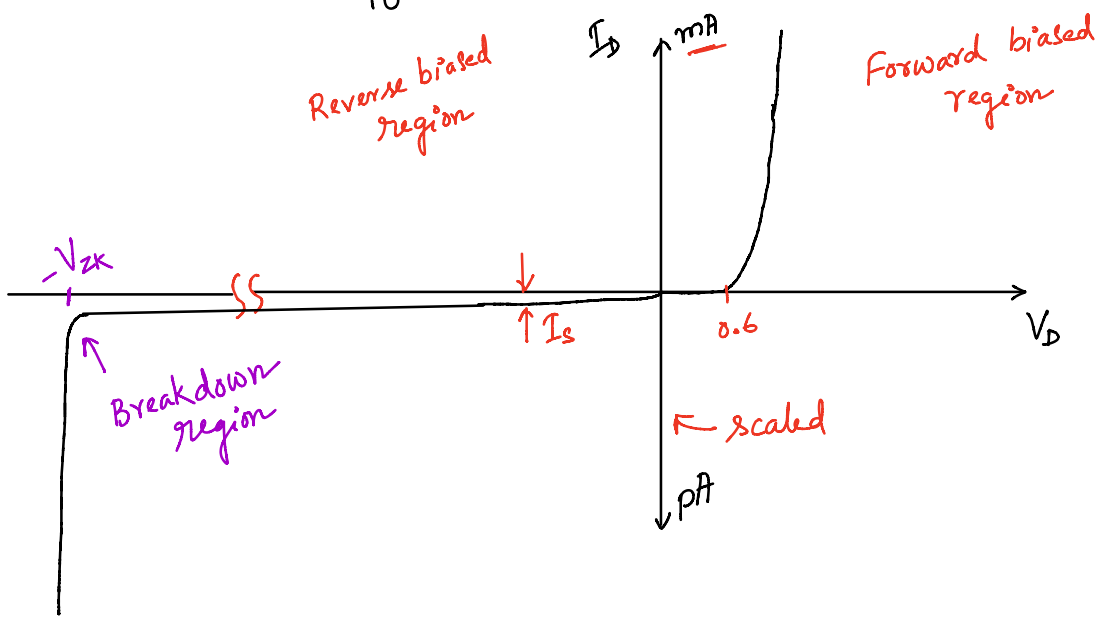
Solving Diode Circuits.





$$I_D = I_S\left(e^{V_D/V_T} - 1\right) \quad \text{where} \quad V_T = \frac{kT}{q}$$

$I_S \rightarrow 10^{-12}\,A$

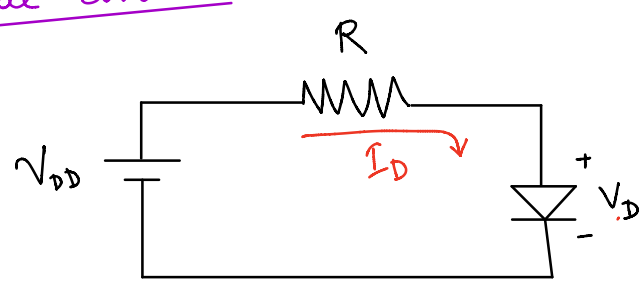


$$I_D = I_S\left(e^{V_D/V_T} - 1\right)$$

$$\Rightarrow I_D = I_S e^{V_D/V_T} - I_S$$

$$\boxed{\therefore I_D \simeq I_S e^{V_D/V_T}}$$

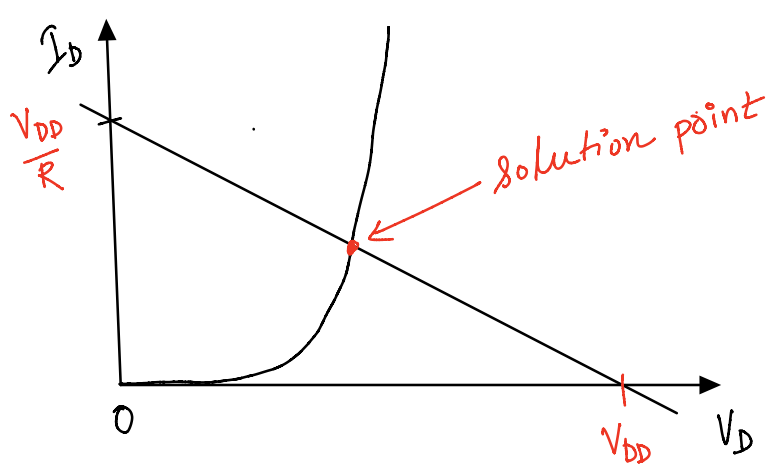
Solving diode circuit



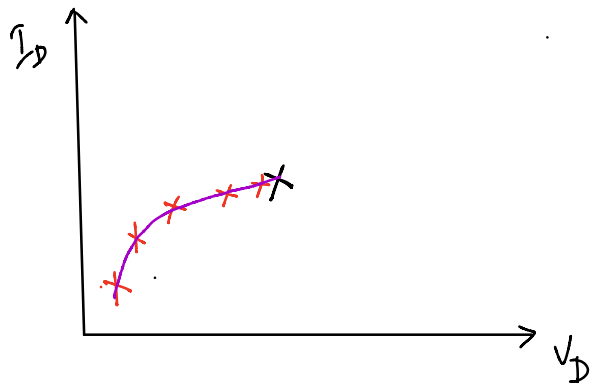
$$I_D = I_S e^{V_D/V_T} \longrightarrow (i)$$

$$I_D = \frac{V_{DD} - V_D}{R} \longrightarrow (ii)$$

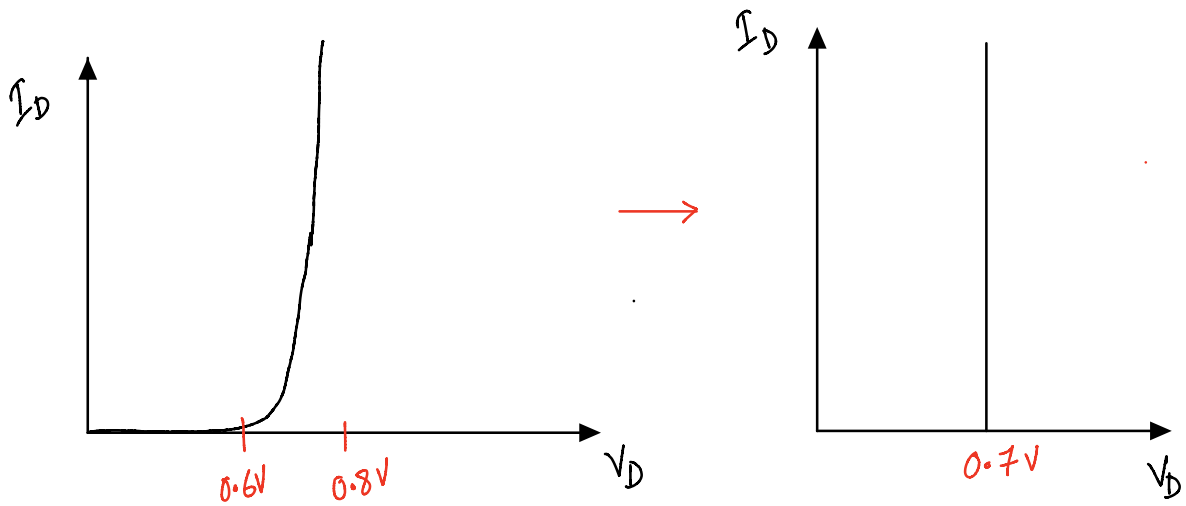
Graphical method

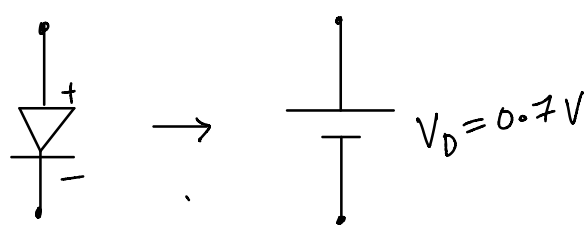


Numerical Method / Iterative Method

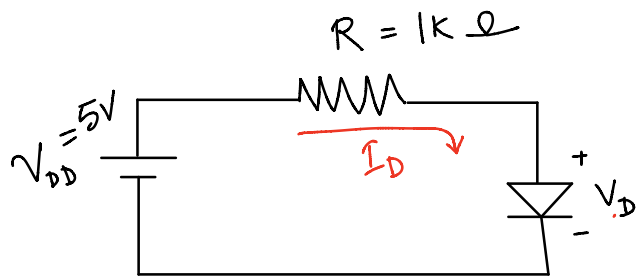


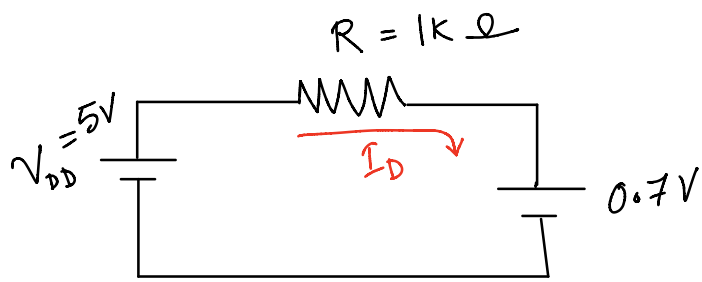
Constant Drop Model





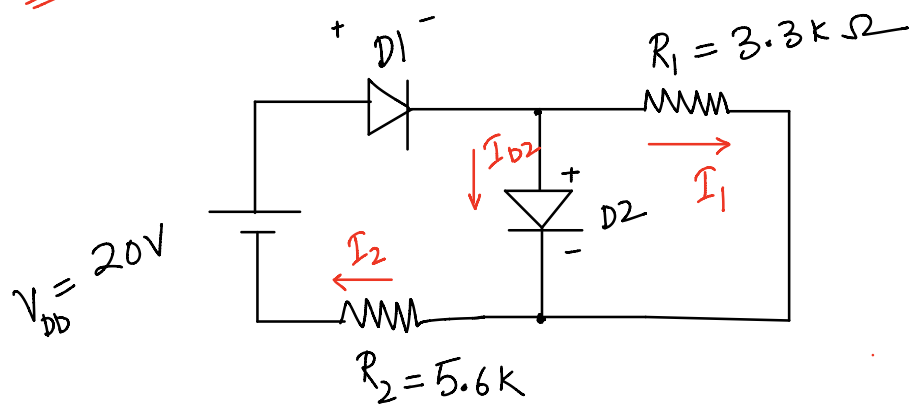
Q1: Find I_D in the following example





$$\therefore I_D = \frac{V_{DD} - 0.7}{R} = \frac{5 - 0.7}{1K} = 4.3mA$$

Q2: Find I_1, I_2 and I_{D2}



$$I_1 R_1 = 0.7\ V \Rightarrow I_1 = \frac{0.7}{3.3K} = 0.212mA$$

$$\therefore V_{DD} - V_{D1} - V_{D2} - I_2 R_2 = 0$$

$$\Rightarrow I_2 = \frac{V_{DD} - V_{D1} - V_{D2}}{R_2} = \frac{20 - 0.7 - 0.7}{5.6K}$$

$$= 3.32\ mA$$

$$\therefore I_2 = I_{D2} + I_1$$

$$\Rightarrow I_{D2} = I_2 - I_1 = 3.11mA$$

$I_1 = 0.212mA$ $\qquad$ $I_{D2} = 3.11mA$

$I_2 = 3.32mA$